

Technical Jargon & System Terminology Reviewer

Master Technical Keywords, Architecture Terms & Defense Defense Vocabulary

Target System: CDM Integrated Library Management System (CDM ILMS)

Scope: Library Standards, React Architecture, REST API, SQLite & Security

Purpose: Master Defense Jargon & Panel Terminology

Format: Technical Term + Code Context + Taglish Explanation

1. Academic Library & Cataloging Terminology

MARC 21 Tagging Standard (Machine-Readable Cataloging)

An international metadata standard used by academic libraries to structure bibliographic data so computers can read, exchange, and index catalog records. Key tags used in CDM ILMS: Tag 001 (Control No.), Tag 020 (ISBN), Tag 082 (Dewey Classification), Tag 100 (Main Author), and Tag 245 (Title Statement).

Paliwanag sa Defense: Ito ang standard format na ginagamit natin sa pag-store ng information ng libro para structured at madaling i-search or i-export sa ibang academic library systems.

Library of Congress Call Number (LC Call No.)

A unique alphanumeric code assigned to each book catalog entry (e.g., QA 76.73.J38 S26 2024) indicating its exact subject classification and physical shelf position in the library.

Paliwanag sa Defense: Ito 'yung exact address o location badge ng libro sa bookshelf batay sa subject area nito katulad ng Computer Science (QA) o Business (HF).

Circulation Ledger

The automated real-time transaction ledger that tracks book check-outs, issue dates, return due dates, active loans, and overdue calculations.

Paliwanag sa Defense: Ito ang digital record ng lahat ng hiram at balik ng libro sa system natin, kasama ang pag-calculate kung overdue na ang hiram ng estudyante.

Accessioning & Copy Inventory Control

The process of tracking available stock count (`available`) versus total owned physical copies (`total`) of each reference volume in the database.

Paliwanag sa Defense: Ito ang feature kung saan kinokontrol ng system ang count ng available na libro sa shelf para hindi ma-overborrow ang stock.

2. Front-End UI/UX & React Architecture Terminology

Single-Page Application (SPA)

A web app architecture where the entire desktop client loads inside a single web document (`index.html`) and updates views dynamically via React component state without full page reloads.

Paliwanag sa Defense: *Ibig sabihin hindi nagre-refresh ang buong browser tuwing naglilipat tayo ng page katulad ng Dashboard o Catalog, kaya napakabilis ng navigation.*

Progressive Web App (PWA)

A modern web application framework that utilizes Service Workers and Web Manifest (`manifest.webmanifest`) to allow mobile users to install the system on smartphones and view cached catalog data offline.

Paliwanag sa Defense: *Ito 'yung mobile application natin na pwedeng i-install sa cellphone ng estudyante kahit walang Google Play Store o App Store.*

Memoized Search & Filtering (`useMemo`)

A React performance optimization hook that caches filtered and sorted book search results, re-executing only when search keywords or institute filter states change.

Paliwanag sa Defense: *Feature ito na nagse-save ng search result sa memory para maging smooth at fast (60 FPS) ang pag-filter kahit libo-libo ang libro.*

Stateful Fallback Cover Rendering (`imageError`)

A UI resilience pattern in `BookCard` where missing or broken image cover URLs automatically transition to a typographic book spine card without throwing visual errors.

Paliwanag sa Defense: *Kapag sira o kulang ang picture URL ng libro, hindi mag-e-error ang screen; kusa itong magdi-display ng magandang typographic spine card.*

Segmented Control Pill Group

A modern segmented UI filter button container for switching between academic institutes (`All`, `ICS`, `ITE`, `IBE`) with real-time book count badges.

Paliwanag sa Defense: *Ito 'yung sleek na pill-button bar sa taas ng Catalog kung saan pumipili ang librarian ng institute para mag-filter.*

Keyboard Focus Shortcut (`K` / `Ctrl+K`)

An intuitive accessibility feature that maps global keyboard events to immediately place focus on the catalog search bar.

Paliwanag sa Defense: *Pindutin lang ang Ctrl+K o Cmd+K sa keyboard, kusa nang magpi-focus ang cursor sa search bar para mabilis na paghahanap.*

3. Backend REST API, Security & Database Terminology

RESTful REST API (Representational State Transfer)

An architectural style for web services using standard HTTP verbs (`GET`, `POST`, `PUT`, `DELETE`) and structured JSON payloads to communicate between frontend and backend.

Paliwanag sa Defense: *Ito ang tulay sa pagitan ng React UI natin at ng Server: nagpapadala at kumukuha tayo ng data gamit ang HTTP JSON requests.*

Stateless JWT Authentication (JSON Web Token)

A secure token-based session verification protocol. Upon login, the server issues a signed cryptographic token sent in the `Authorization: Bearer` header.

Paliwanag sa Defense: *Naka-sign na digital pass ito na binibigay ng server pagka-login ng librarian, para ma-verify kung authorized siyang mag-access ng confidential endpoints.*

Password Salting & Hashing (`bcrypt`)

A password security algorithm that applies random salt strings and 10 calculation iterations to hash passwords before storing them in SQLite.

Paliwanag sa Defense: *Hinding-hindi natin itinatago nang plain-text ang password sa database. Kino-convert ito ng `bcrypt` sa un-readable hash value.*

Email OTP Recovery Engine (One-Time Password)

An automated account recovery flow that generates a 6-digit numeric OTP code stored in SQLite (`otp\_codes` table) with a 10-minute expiration timestamp.

Paliwanag sa Defense: *Kapag nakalimutan ng user ang password, nagpapadala ang system ng 6-digit PIN sa email na may 10 minutes validation limit.*

Synchronous Embedded Database (`better-sqlite3`)

A zero-config, embedded relational database driver for SQLite in Node.js that executes SQL statements synchronously without async promise overhead.

Paliwanag sa Defense: *Ito ang high-performance database driver natin na napakabilis magbasa at mag-write ng data para sa library transactions.*

Relational Schema & Foreign Key Integrity

Database structural design enforcing relational links between tables (e.g., `transactions.bookId` referencing `books.id`), preventing orphaned records.

Paliwanag sa Defense: *Sinisigurado nito na bawat hiram sa database ay totoong nakakonekta sa tamang libro at tamang estudyante.*

4. Hardware & Identity Verification Terminology

2D QR Receipt Canvas Generation (`qrcode`)

Client-side dynamic 2D barcode canvas rendering that encodes verified transaction payloads and student IDs into scannable QR codes.

Paliwanag sa Defense: *Kusa itong nagse-generate ng 2D QR Code image sa screen matapos maghiram ang estudyante para sa turnstile verification.*

Turnstile Exit Verification Checkpoint

A physical or digital security verification step where library staff or scanners scan the student's 2D QR receipt to confirm authorized book exit.

Paliwanag sa Defense: *Ito ang security step sa pinto ng library kung saan kino-confirm sa QR Code kung legal na na-borrow ang librong ilalabas.*